المدرسة الوطنية للعلوم التطبيقية - بني ملال

ⵜⴰⵙⴷⴰⵡⵉⵜ ⵏ ⵜⵎⵓⵙⵙⵓⵏⵉⵏ ⵜⵉⵎⵓⵔⵉⵏ - ⴱⵏⵉ ⵎⵍⴰⵍ

Ecole Nationale des Sciences Appliquées - Béni Mellal

Informations Générales

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- **Encadre par** : Pr : B.ELBAGHAZAOUI

TP4 : Architecture Orientée Service (SOA)

Objectifs :

• Comprendre les principes de l'architecture SOA

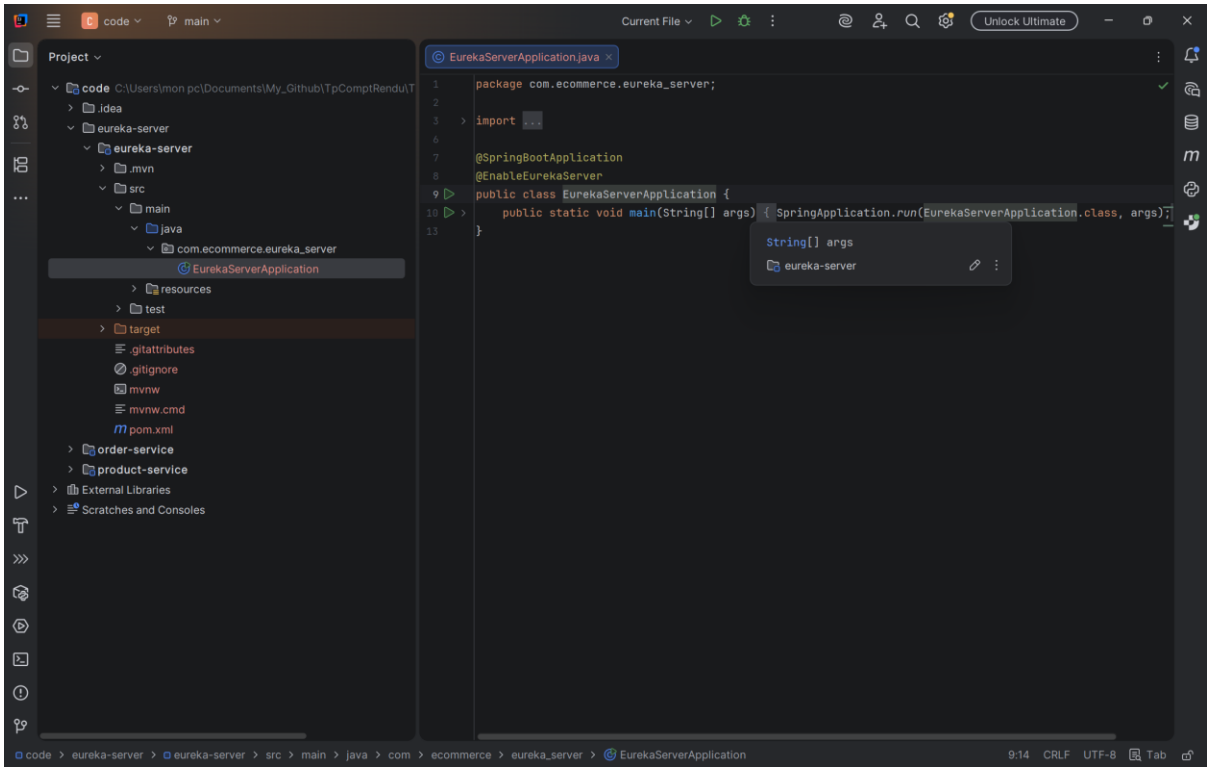
• Créer des microservices avec Spring Boot

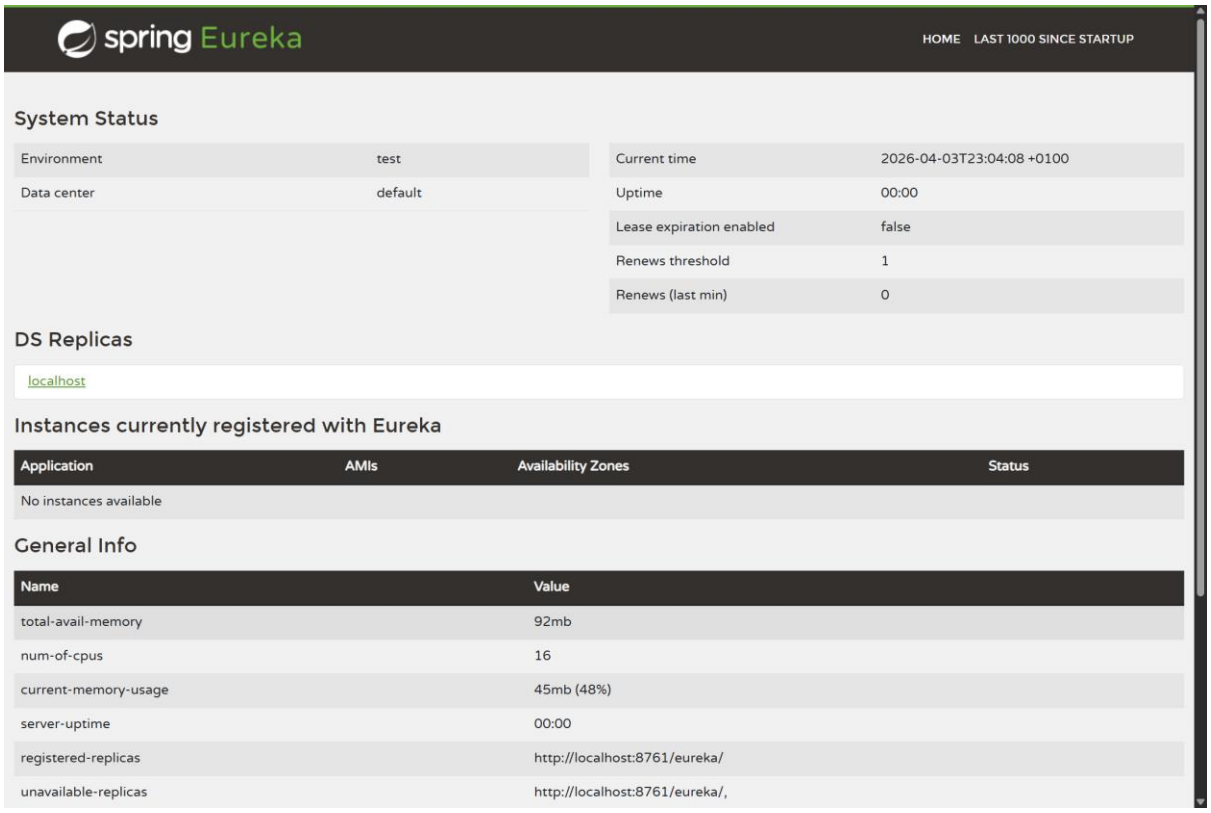
• Implémenter la communication inter-services via REST

• Gérer une base de données avec Spring Data JPA

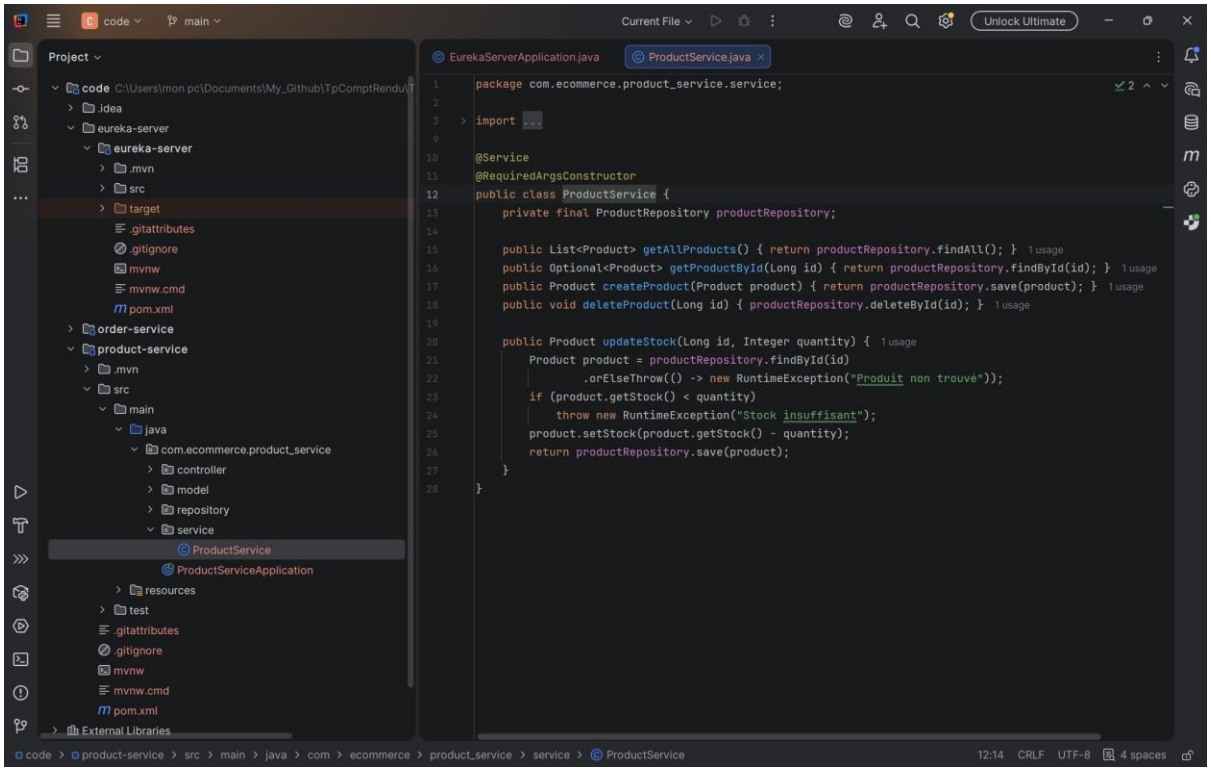
• Utiliser un service de découverte (Eureka)

Etape 1 :

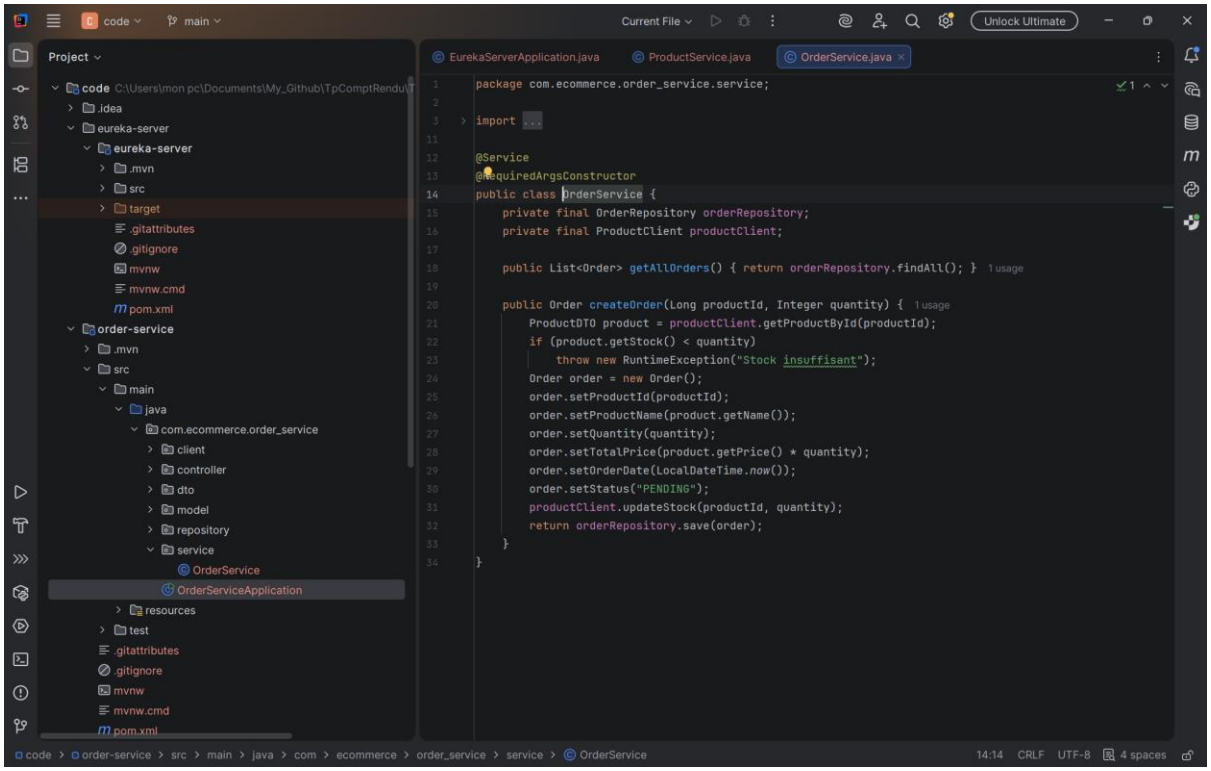

code
main
Current File
Unlock Ultimate
Project
code C:\Users\mon pc\Documents\My_Github\TpComptRendu\T
.idea
eureka-server
eureka-server
.mvn
src
main
java
com.ecommerce.eureka_server
EurekaServerApplication
resources
test
target
.gitattributes
.gitignore
mvnw
mvnw.cmd
pom.xml
order-service
product-service
External Libraries
Scratches and Consoles
EurekaServerApplication.java
package com.ecommerce.eureka_server;
import ...
@SpringBootApplication
@EnableEurekaServer
public class EurekaServerApplication {
public static void main(String[] args) { SpringApplication.run(EurekaServerApplication.class, args);
}
String[] args
eureka-server
code > eureka-server > eureka-server > src > main > java > com > ecommerce > eureka_server > EurekaServerApplication
9:14
CRLF
UTF-8
Tab


spring Eureka
HOME
LAST 1000 SINCE STARTUP
System Status
Environment
test
Data center
default
Current time
2026-04-03T23:04:08 +0100
Uptime
00:00
Lease expiration enabled
false
Renews threshold
1
Renews (last min)
0
DS Replicas
localhost
Instances currently registered with Eureka
Application
AMIs
Availability Zones
Status
No instances available
General Info
Name
Value
total-avail-memory
92mb
num-of-cpus
16
current-memory-usage
45mb (48%)
server-uptime
00:00
registered-replicas
http://localhost:8761/eureka/
unavailable-replicas
http://localhost:8761/eureka/,

Etape 2 :

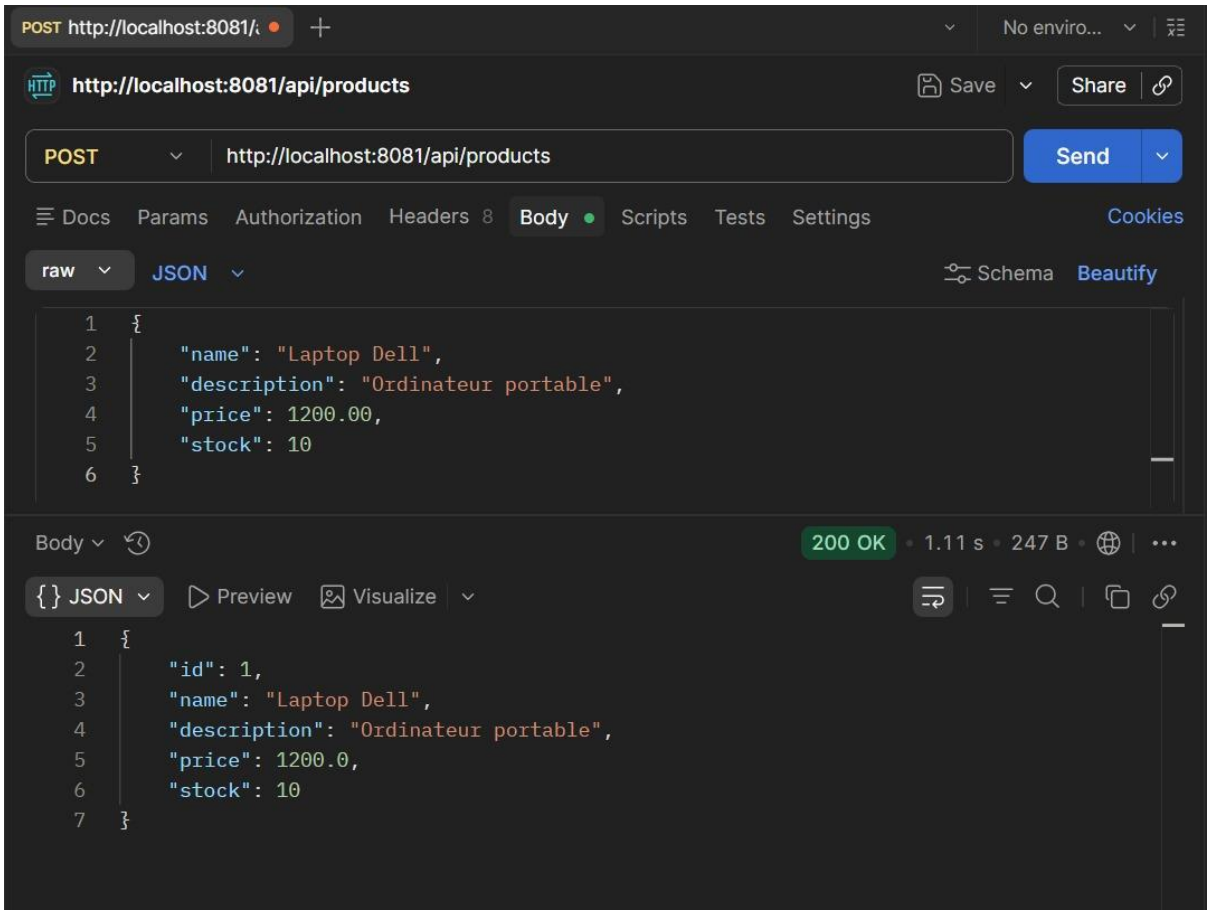

code main
Current File
Unlock Ultimate
Project
code C:\Users\mon pc\Documents\My_Github\TpComptRendu\T
.idea
eureka-server
eureka-server
.mvn
src
target
.gitattributes
.gitignore
mvnw
mvnw.cmd
pom.xml
order-service
product-service
.mvn
src
main
java
com.ecommerce.product_service
controller
model
repository
service
ProductService
ProductServiceApplication
resources
test
.gitattributes
.gitignore
mvnw
mvnw.cmd
pom.xml
External Libraries
EurekaServerApplication.java
ProductService.java
package com.ecommerce.product_service.service;
import ...
@Service
@RequiredArgsConstructor
public class ProductService {
private final ProductRepository productRepository;
public List<Product> getAllProducts() { return productRepository.findAll(); } 1 usage
public Optional<Product> getProductById(Long id) { return productRepository.findById(id); } 1 usage
public Product createProduct(Product product) { return productRepository.save(product); } 1 usage
public void deleteProduct(Long id) { productRepository.deleteById(id); } 1 usage
public Product updateStock(Long id, Integer quantity) { 1 usage
Product product = productRepository.findById(id)
.orElseThrow(() -> new RuntimeException("Produit non trouvé"));
if (product.getStock() < quantity)
throw new RuntimeException("Stock insuffisant");
product.setStock(product.getStock() - quantity);
return productRepository.save(product);
}
}
code > product-service > src > main > java > com > ecommerce > product_service > service > ProductService
12:14 CRLF UTF-8 4 spaces

Etape 3 :

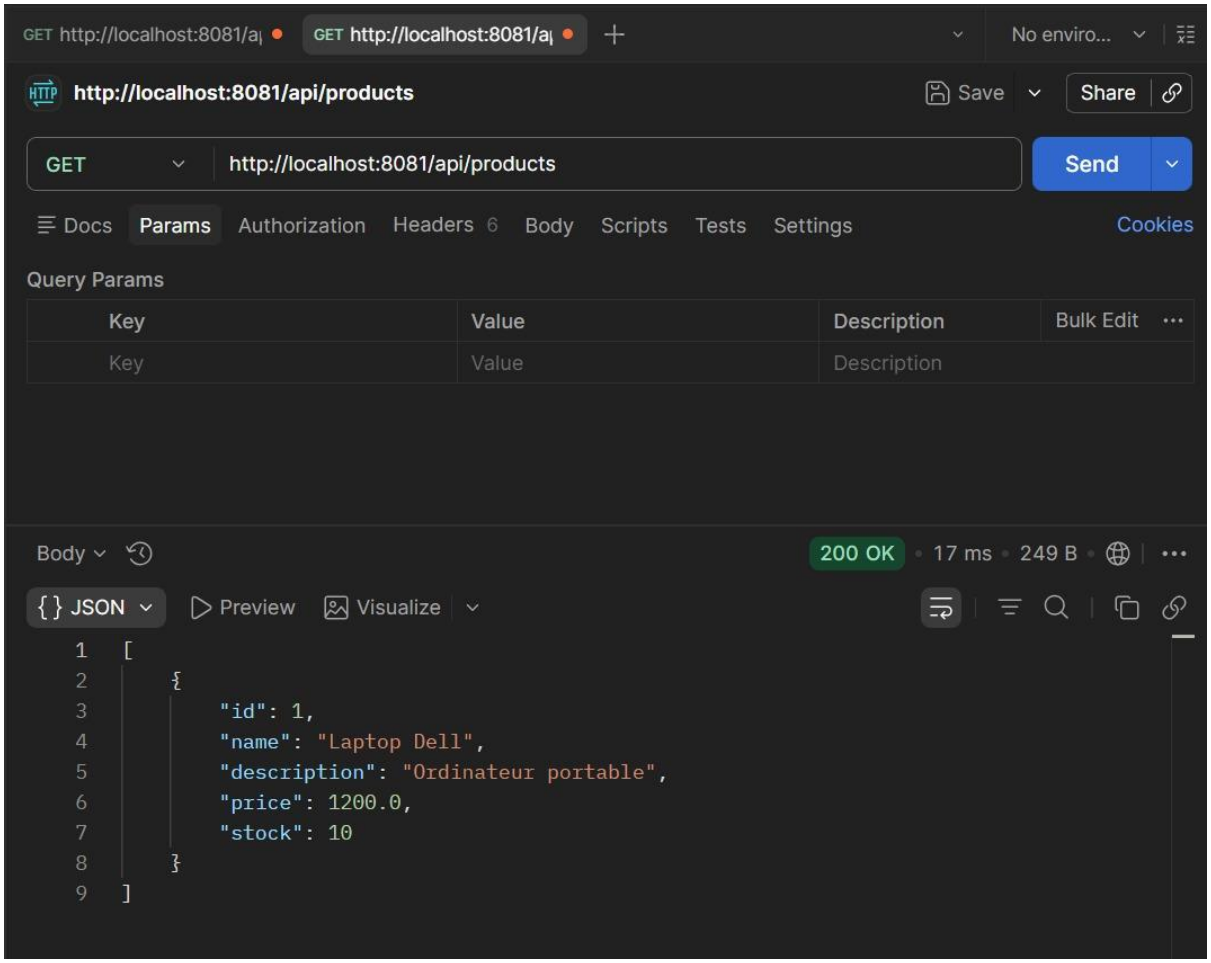

code main
Current File
Unlock Ultimate
Project
code C:\Users\mon pc\Documents\My_Github\TpComptRendu\T
.idea
eureka-server
eureka-server
.mvn
src
target
.gitattributes
.gitignore
mvnw
mvnw.cmd
pom.xml
order-service
.mvn
src
main
java
com.ecommerce.order_service
client
controller
dto
model
repository
service
OrderService
OrderServiceApplication
resources
test
.gitattributes
.gitignore
mvnw
mvnw.cmd
pom.xml
EurekaServerApplication.java
ProductService.java
OrderService.java
package com.ecommerce.order_service.service;
import ...
@Service
@RequiredArgsConstructor
public class OrderService {
private final OrderRepository orderRepository;
private final ProductClient productClient;
public List<Order> getAllOrders() { return orderRepository.findAll(); } 1 usage
public Order createOrder(Long productId, Integer quantity) { 1 usage
ProductDTO product = productClient.getProductById(productId);
if (product.getStock() < quantity)
throw new RuntimeException("Stock insuffisant");
Order order = new Order();
order.setProductId(productId);
order.setProductName(product.getName());
order.setQuantity(quantity);
order.setTotalPrice(product.getPrice() * quantity);
order.setOrderDate(LocalDateTime.now());
order.setStatus("PENDING");
productClient.updateStock(productId, quantity);
return orderRepository.save(order);
}
}
code > order-service > src > main > java > com > ecommerce > order_service > service > OrderService
14:14 CRLF UTF-8 4 spaces

Tests :

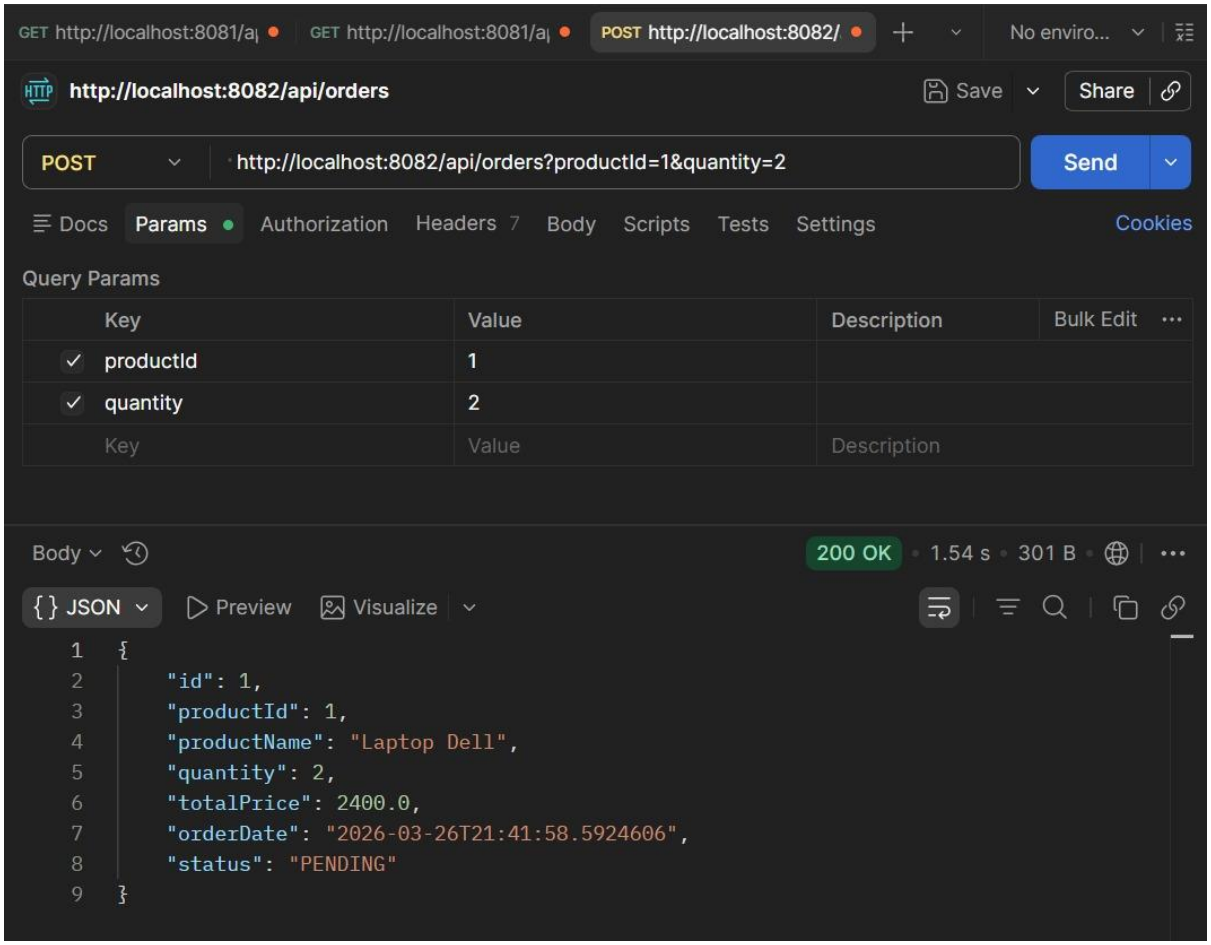
Test 1 :



Test 2 :


GET http://localhost:8081/a
GET http://localhost:8081/a
No enviro...
http://localhost:8081/api/products
Save
Share
GET
http://localhost:8081/api/products
Send
Docs
Params
Authorization
Headers 6
Body
Scripts
Tests
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Key
Value
Description
Body
200 OK
17 ms
249 B
JSON
Preview
Visualize
[
{
"id": 1,
"name": "Laptop Dell",
"description": "Ordinateur portable",
"price": 1200.0,
"stock": 10
}
]

Test 3 :


GET http://localhost:8081/a
GET http://localhost:8081/a
POST http://localhost:8082/
No enviro...
http://localhost:8082/api/orders
Save
Share
POST
http://localhost:8082/api/orders?productId=1&quantity=2
Send
Docs
Params
Authorization
Headers 7
Body
Scripts
Tests
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
productId
1
quantity
2
Key
Value
Description
Body
200 OK
1.54 s
301 B
JSON
Preview
Visualize
{
"id": 1,
"productId": 1,
"productName": "Laptop Dell",
"quantity": 2,
"totalPrice": 2400.0,
"orderDate": "2026-03-26T21:41:58.5924606",
"status": "PENDING"
}

Test 4 :

GET http://localhost | GET http://localhost | POST http://localhos | GET http://localhost | No enviro...

http://localhost:8082/api/orders

Save | Share

GET http://localhost:8082/api/orders Send

Docs | Params | Authorization | Headers 6 | Body | Scripts | Tests | Settings | Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body | 200 OK • 567 ms • 302 B

{ } JSON | Preview | Visualize


```
[
    {
        "id": 1,
        "productId": 1,
        "productName": "Laptop Dell",
        "quantity": 2,
        "totalPrice": 2400.0,
        "orderDate": "2026-03-26T21:41:58.592461",
        "status": "PENDING"
    }
]
```